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CSE Student Hub



<p> An intelligent, self-updating web portal for academic resources, powered by Python and GitHub Actions. </p>

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CSE Student Hub is not just another file-hosting page; it's a fully automated content management system. Simply add a file to the repository, and the website instantly rebuilds itself to include the new resource in the correct category. No more manual HTML editing—ever.



Core Features

- ✨ **Zero-Effort Updates:** Drop a file into the repository, git push, and watch it appear on the site seconds later.
- 🧠 **Smart Categorization:** Uses filename keywords to automatically sort files by course code (CS-501, CS-502) and file type (PDF, EPUB, etc.).
- 🎨 **User Personalization:** A sleek, pop-up profile page lets users set their name and upload a custom avatar, which persists across visits.
- 🌙 **Light & Dark Mode:** A modern theme toggle for comfortable viewing in any lighting condition.
- 🔍 **Instant Search:** A responsive search bar that filters all resources in real-time.
- 📱 **Fully Responsive:** A clean, professional UI that works flawlessly on desktop and mobile devices.



The Automation Engine: How It Works

The project's core is a powerful feedback loop between your file system and the live website, orchestrated entirely by GitHub Actions.

graph TD

A[👤 You: git push new file] --> B{🔄 GitHub Actions Workflow Triggered};

B --> C[🐍 Python Script: generate_index.py];

C --> D{1. Scan Directory & Collect Files};

D --> E{2. Categorize by Course & Type};

E --> F{3. Generate HTML Content};

F --> G[📄 Overwrite index.html];

G --> H{🚀 Commit Updated index.html};

```
H --> I[🌐 Website is Live & Updated!];
```

```
style A fill:#7C3AED,stroke:#fff,stroke-width:2px,color:#fff
```

```
style I fill:#22c55e,stroke:#fff,stroke-width:2px,color:#fff
```

1. **The Trigger:** When you git push, a workflow defined in `.github/workflows/` kicks in.
2. **The Engine:** It runs the `generate_index.py` script.
3. **The Logic:** The script scans for all valid files, intelligently sorts them, builds the necessary HTML for the tabs and file cards, and injects it all into a master template.
4. **The Result:** The script saves a brand new `index.html` file.
5. **The Update:** The workflow commits this new `index.html` back to the repository, and GitHub Pages automatically deploys the updated version.

Technology Stack

Technology	Purpose
Python	Core automation logic, file scanning, and HTML generation.
GitHub Actions	CI/CD pipeline for triggering the Python script on every push.
HTML5	The structural backbone of the web portal.
CSS3	Custom styling, responsive design, themes, and the profile modal.
JavaScript	Powers all client-side interactivity, including search, tabs, and user profile management.

How to Add New Files

Managing the site content is as simple as managing a folder.

1. **Add Your File:** Place your new document (e.g., `Cyber_Security_Notes_Unit_3.pdf`) into the root of the project folder. **Tip:** Name your files descriptively! The script uses keywords like `cyber`, `dbms`, `toc`, etc., to auto-categorize.
2. **Commit and Push:** Use the standard git commands.

```
git add .
```

```
git commit -m "feat: Add cyber security unit 3 notes"
```

```
git push
```